

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Bioscience and Biotechnology

2. **Subject Name:** Bioinformatics

L-T-P: 3-1-0

Credits: 4

3. **Pre-requisites:** Biochemistry (BT11002)

4. **Syllabus and reference books:**

Syllabus:

1. Introduction to Bioinformatics: Key concepts of macromolecular sequences, structures and their functions.
2. Understanding Databases: Introduction to Sequence and Structure databases of Proteins and Nucleic acids of different organisms. Data handling in PDB, Detailed overview of NCBI Entrez
3. Sequence alignment: Algorithms for Local and Global sequence alignments, Pairwise and Multiple sequence alignments, BLAST analysis of nucleotide and Protein sequences.
4. Prediction of Secondary and tertiary structures of proteins, Homology modeling, Threading and Ab-initio modeling, fold recognition, overview of CASP
5. Structure alignment: Algorithms for macromolecular structure alignments, Introduction of DALI
6. Macromolecular visualization tools: using Rasmol, PyMol and VMD
7. Structural classification of proteins: Alpha, beta and alpha/beta class of proteins, Introductions to CATH and SCOP databases
8. Evolution of macromolecular sequences and structures: Basis of molecular phylogeny, Phylogenetic tree construction and interpretations.
9. Introduction to Next Generation Sequencing Technologies
10. NGS data formats and Quality control
11. Read mapping algorithms and tools
12. SAM and BAM files, Variant calling, CNV analysis
13. RNA-sequencing data analysis - Data normalization, Differential gene expression analysis, Multiple hypothesis testing and Gene Ontology (GO)
14. De-novo assembly algorithms

Reference Books:

- 1) Bioinformatics: Sequence and Genome analysis David W. Mount
- 2) Introduction to Protein Structure, Carl Branden and John Tooze
- 3) Structural Bioinformatics by Jenny Gu and Philip E. Bourne
- 4) Bioinformatics and Functional Genomics by J. Pevsner
- 5) High-Throughput Next Generation Sequencing, Methods and Applications. (Springer). Editors: Kwon, Young Min, Ricke, Steven C. (Eds.)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to Bioinformatics: Key concepts of macromolecular sequences, structures and their functions.	2
2.	Understanding Databases: Introduction to Sequence and Structure databases of Proteins and Nucleic acids of different organisms. Data handling in PDB, Detailed overview of NCBI Entrez	4
3.	Sequence alignment: Algorithms for Local and Global sequence alignments, Pairwise and Multiple sequence alignments, BLAST analysis of nucleotide and Protein sequences.	4
4.	Prediction of Secondary and tertiary structures of proteins, Homology modeling, Threading and Ab-initio modeling, fold recognition, overview of CASP	5
5.	Structure alignment: Algorithms for macromolecular structure alignments, Introduction of DALI	3
6.	Macromolecular visualization tools: using Rasmol, PyMol and VMD	3
7.	Structural classification of proteins: Alpha, beta and alpha/beta class of proteins, Introductions to CATH and SCOP databases	5
8.	Evolution of macromolecular sequences and structures: Basis of molecular phylogeny, Phylogenetic tree construction and interpretations.	2
9.	Introduction to Next Generation Sequencing Technologies	3
10.	NGS data formats and Quality control	2
11.	Read mapping algorithms and tools	4
12.	SAM and BAM files, Variant calling, CNV analysis	4
13.	RNA-sequencing data analysis - Data normalization, Differential gene expression analysis, Multiple hypothesis testing and Gene Ontology (GO)	6
14.	De-novo assembly algorithms	3
Total number of hours		50